

- Q.19 Explain the principle of Vat Photo-polymerization technique in additive manufacturing. (CO4)
- Q.20 What are the limitations of materials extrusion technique? (CO4)
- Q.21 Discuss any four defects and their causes in 3D printing. (CO5)
- Q.22 Explain Wire-cut EDM? (CO5)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain Additive Manufacturing chain. Also enlist the applications of AM. (CO3)
- Q.24 Describe the principle, technologies, and potential benefits of Powder Bed Fusion in additive manufacturing. (CO4)
- Q.25 Write /Short notes on (CO5)
- Visual Inspection
 - Surface roughness measurement

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3rd Sem / Automation & Robotics

Subject : AutoCAD and 3D Printing

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 What is 3D printing? (CO3)
- Creating two-dimensional designs
 - Adding layers to create three-dimensional objects
 - Printing text documents on paper
 - Printing holographic images
- Q.2 Which additive manufacturing method uses a laser to solidify powdered materials? (CO4)
- Fused deposition modeling (FDM)
 - Stereolithography (SLA)
 - Selective laser sintering (SLS)
 - Digital light processing (DLP)

- Q.3 AutoCAD commands can be accessed in (CO1)
 a) Drawing area b) Status bar
 c) Tool bars d) Command window
- Q.4 In AutoCAD, the number of workspaces is (CO2)
 a) 1 b) 2
 c) 3 d) 4
- Q.5 Which of the following types of material does the material jetting technology use? (CO5)
 a) Wax b) Glass
 c) Photopolymer d) Foam
- Q.6 What are the two distinct 3-dimensional coordinate systems used by AutoCAD? (CO1)
 a) The WCC and UCS. b) The WCS and UCC.
 c) The WSC and USC. d) The WCS and UCS.

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 STL stands for stereolithography. (T/F) (CO3)
- Q.8 Full form of UCS is _____ (CO1)

- Q.9 Additive manufacturing uses much more material than other subtractive manufacturing processes. (T/F) (CO3)
- Q.10 Write the shortcut command for making polylines. (CO1)
- Q.11 Write any two applications of 3D printing. (CO4)
- Q.12 Write name of commercial 3D printing software. (CO4)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Explain the coordinate systems used in AutoCAD. (CO1)
- Q.14 Explain the procedure for opening a new file in AutoCAD. (CO2)
- Q.15 What are the benefits of additive manufacturing over conventional manufacturing? (CO3)
- Q.16 List the different features of 3D printing softwares. (CO3)
- Q.17 Describe the basic steps involved in additive manufacturing (AM) from uploading CAD files to producing a 3D-printing object. (CO3)
- Q.18 Explain key parameters involved in Direct Energy Deposition method. (CO4)